Python Programming Lab. [Batch-B3]

Assignment Submission Details

Field	Details
Student Name *	Dilip Balu Magar
Student PRN *	125M1H055
Course Name	Python Programming Lab. (MCA31PC06)
Academic Year	2025-26 (Semester-1)
Course Teacher	Prof. Prakash Ukhalkar
Assignment Name	Assignment based on Functions in Python
Assignment Number	Assignment 04
Submission Date *	17-11-2025

Instructions

1. **Fill in your details** in the table above
2. **Write your code** in the provided code cells below each question
3. **Test your code** to ensure it works correctly
4. **Add comments** to explain your logic
5. **Save the notebook** before submission

Question 1: Python Program to Find Numbers Divisible by Another Number.

In [ ]:
```
def find_div_nums(num_list, divisor):
    new_list = [num for num in num_list if num % divisor == 0]
    #new_list stores numbers which are divisible by divisor from num_list
    return new_list


num_list = []

n= int(input("enter no elements to add in list"))

for i in range(0,n):
    num=int(input(f"enter element {i}"))
    num_list.append(num)  #add numbers in list
```

```
divisor=int(input("enter divisor:"))

print("Given list",num_list)
print(f"numbers from list divisible by {divisor} are:{find_div_nums(num_list,diviso
```

```
Given list [5, 10, 15, 3, 4]
numbers from list divisible by 5 are:[5, 10, 15]
```

Question 2:Python Program to Convert Decimal to Binary, Octal, and Hexadecimal.

In [11]:
```
decimal=int(input("enter the number"))

binary=bin(decimal)
octal=oct(decimal)
hexa=hex(decimal)

print(f"given decimal number = {decimal}")
print(f"binary conversion of decimal number = {binary[2:]}")
print(f"ocatl conversion of decimal number = {octal[2:]}")
print(f"hexadecimal conversion of decimal number = {hexa[2:]}")
#we use [2:] to remove the prefix eg 0b1010 to 1010
```

```
given decimal number = 10
binary conversion of decimal number = 1010
ocatl conversion of decimal number = 12
hexadecimal conversion of decimal number = a
```

Question 3:Python Program to Find ASCII Value of Character.

In [13]:
```
char=input("enter a single character to find its ascii value")

if len(char) > 1:
    print( "enter single character")
else:
    print(f"ascii value of {char} is {ord(char)}")
```

```
ascii value of A is 65
```

Question 4: Python Program to Find the H.C.F and L.C.M of Two input Numbers

In [ ]:
```
def cal_hcf(x, y):


    smaller = min(x, y) #as a number cannt have  factor greater than itself
    hcf = 1             #minimum possible HCF
    for i in range(1, smaller + 1):  #including the smaller number.
        if (x % i == 0) and (y % i == 0):
            hcf = i      # i is a common factor
    return hcf

def cal_lcm(x, y):

    greater = max(x, y)
```

```
    lcm = greater  # LCM must be at least the greater number.

    # WE CAN ALSO USE THIS TO FIND :LCM = (n1*n2)/HCF

    while True:    #until LCM found
        if (lcm % x == 0) and (lcm % y == 0):
            return lcm
        lcm += greater

n1 = int(input("enter a number 1 to find LCM and HCF"))
n2 = int(input("enter a number 2 to find LCM and HCF"))
hcf_result = cal_hcf(n1, n2)
lcm_result = cal_lcm(n1, n2)

print(f"The H.C.F. of {n1} and {n2} is: {hcf_result}")
print(f"The L.C.M. of {n1} and {n2} is: {lcm_result}")
```

```
The H.C.F. of 15 and 35 is: 5
The L.C.M. of 15 and 35 is: 105
```

Question 5:Python Program to Find Factors of Number

In [9]:
```
def factors(number):
    if number <= 0:
        return "enter positive integer."

    factors = []

    for i in range(1, number + 1):  #Factors include 1 and number itself
        if number % i == 0:
            factors.append(i)
    return f"The factors of {number} are: {factors}"

n =int(input("enter a number to find its factor:"))
print(factors(n))
```

```
The factors of 50 are: [1, 2, 5, 10, 25, 50]
```

Question 6: Python Program makes a simple calculator that can add,subtract,multiply and divide using functions

In [10]:
```
def add(x, y):
    return x + y

def subtract(x, y):
    return x - y

def multiply(x, y):
    return x * y

def divide(x, y):
    if y == 0:
        return "Error: Cannot divide by zero"
    return x / y
```

```
a=int(input("enter number1:"))
b=int(input("enter number2:"))
addition = add(a, b)
print(f"The addition of {a} and {b} is: {addition}")

sub = subtract(a, b)
print(f"The difference between {a} and {b} is: {sub}")

mul = multiply(a, b)
print(f"The product of {a} and {b} is: {mul}")

div = divide(a, b)
print(f"The quotient of {a} divided by {b} is: {div}")
```

```
The addition of 5 and 10 is: 15
The difference between 5 and 10 is: -5
The product of 5 and 10 is: 50
The quotient of 5 divided by 10 is: 0.5
```

Question 7:Python Program to shuffle a deck of cards using the module random and draw 5 Cards

In [ ]:

```
import random

def deck():

    suits = ['Hearts', 'Diamonds', 'Clubs', 'Spades']
    ranks = ['2', '3', '4', '5', '6', '7', '8', '9', '10', 'Jack', 'Queen', 'King',
    deck = [rank + ' of ' + suit for suit in suits for rank in ranks]

    # deck = []
    # for suit in suits:
    #     for rank in ranks:
    #         card_name = rank + ' of ' + suit
    #         deck.append(card_name)
    # return deck

    return deck

def shuffle_and_draw(deck, cards_to_draw=5):

    shuffled_deck = deck[:] #Make a copy to keep original as it is
    random.shuffle(shuffled_deck)

    drawn_cards = shuffled_deck[:cards_to_draw]
    return drawn_cards

full_deck = deck()
drawn_cards = shuffle_and_draw(full_deck, 5)

print(f"Total cards in the deck: {len(full_deck)}")
print("\nYour drawn 5 cards:")
for card in drawn_cards:
    print(f" {card}")
```

```
Total cards in the deck: 52

Your drawn 5 cards:
 King of Hearts
 Jack of Spades
 4 of Clubs
 10 of Diamonds
 2 of Diamonds
```

Question 8:Python Program to display a calendar of a given month of the year

In [ ]:
```python
import calendar

def display(year, month):
    if year<0 or month>12 or month < 0:
        return "please provide valid year and month"

    cal = calendar.month(year, month)

    # month_name is list which contain all month names
    print(f"\nCalendar for {calendar.month_name[month]} {year}")
    print("-" * 35)
    print(cal)


year = int(input("Please enter the Year >0 (e.g., 2025): "))

month = int(input("Please enter the Month > 0 and < 13(1=Jan, 12=Dec): "))


display(year, month)
```

```
Calendar for November 2025
-----------------------------------
   November 2025
Mo Tu We Th Fr Sa Su
                 1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
```

Question 9:Python Program to fnd sum of natural numbers up to n using recursive function

In [25]:
```python
def sum(n):
    if n <= 0:
        return 0
    elif n == 1:
        return 1
    else:
        return n + sum(n - 1)

n = int(input("enter value of n"))
```

```
result = sum(n)
print(f"The sum of natural numbers up to {n} is: {result}")
```

```
The sum of natural numbers up to 50 is: 1275
```

Question 10:Write a Program to demonstrate all the built-in functions from the Math module

In [33]:
```
import math

def math_functions(x):

    print(f"  Pi  is  {math.pi}")

    print('=' * 50)

    x1=x+0.7
    print(f" The Absolute Value of {x1} is: {math.fabs(x1)}")
    print(f" The Floor value of {x1} is: {math.floor(x1)}")
    print(f" The Ceiling value of {x1}  is: {math.ceil(x1)}")

    print('=' * 50)

    print(f" The Square Root of {x} is: {math.sqrt(x)}")

    print('=' * 50)

    print(f" Natural Log (base e) of {x} is: {math.log(x)}")
    print(f" Logarithm Base 10 of {x} is: {math.log10(x)}")

    print('=' * 50)

    base = 2
    power = 3
    print(f" {base} raised to the power of {power}  is: {math.pow(base, power)}")


    print('=' * 50)

    print(f" The Sine of {x} radians is: {math.sin(x)}")
    print(f" The Cosine of {x} radians is: {math.cos(x)}")
    print(f" {x} radians converts to degrees as: {math.degrees(x)}")

    print('=' * 50)

    print(f" Factorial of {x} is: {math.factorial(x)}")


number = int(input("enter a number"))

math_functions(number)
```

```
 Pi  is  3.141592653589793
==================================================
 The Absolute Value of 4.7 is: 4.7
 The Floor value of 4.7 is: 4
 The Ceiling value of 4.7  is: 5
==================================================
 The Square Root of 4 is: 2.0
==================================================
 Natural Log (base e) of 4 is: 1.3862943611198906
 Logarithm Base 10 of 4 is: 0.6020599913279624
==================================================
 2 raised to the power of 3  is: 8.0
==================================================
 The Sine of 4 radians is: -0.7568024953079282
 The Cosine of 4 radians is: -0.6536436208636119
 4 radians converts to degrees as: 229.1831180523293
==================================================
 Factorial of 4 is: 24
```

Submission Checklist

Before submitting, make sure you have completed the following:

- ☐ Filled in all personal details in the header
- ☐ Completed all 10 questions
- ☐ Added appropriate comments to your code
- ☐ Tested all programs to ensure they work correctly
- ☐ Used proper variable names and coding conventions
- ☐ Saved the notebook file (.ipynb)
- ☐ Followed file naming format as PRN_A01_PPLAB_B3.ipynb